

Galala University

Civara

API Contract

A Public Statistical Analysis API for Student Performance Datasets

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Project Name: Civara

Domain: `civara.mahros.dev`

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1. Overview

Civara is a simple public API designed to analyze student performance datasets. The system allows a user to upload a CSV file, sends the dataset to the API, and receives a statistical analysis response.

The API focuses on descriptive statistics, confidence intervals, and hypothesis testing for comparing two independent groups, such as male and female students.

The API does not require authentication and does not depend on a database. Analysis is performed per request, and any persistence may be handled temporarily through application memory or browser cache.

2. Base URL

The production base URL is:

```
https://civara.mahros.dev/api/v1
```

For local development, the base URL may be:

```
http://localhost:3030/api/v1
```

3. Authentication

The API is public.

No authentication token, API key, or user account is required.

4. Data Persistence

The API does not permanently store uploaded datasets.

The expected behavior is:

- The client uploads a dataset.
- The server analyzes the dataset.
- The server returns the analysis result immediately.

- The client may store the result in browser cache or local storage.
- The server may keep temporary in-memory data only when needed.

5. Supported File Format

The API accepts CSV files.

The uploaded CSV file should contain the following columns:

Column Name	Description
gender	The group variable used for comparison.
math score	Student score in mathematics.
reading score	Student score in reading.
writing score	Student score in writing.

Table 1: Required CSV columns

The API calculates the average score using:

$$\text{Average Score} = \frac{\text{Math Score} + \text{Reading Score} + \text{Writing Score}}{3}$$

6. API Endpoints

7. Standard API Response Envelope

All endpoints in this API return a consistent JSON envelope.

7.1. Success Response (Single Object)

```
{
  "success": true,
  "message": "Request completed successfully",
  "data": {
    // ...
  },
  "meta": null
}
```

7.2. Success Response (List + Pagination)

When an endpoint returns a list, it uses the same envelope, and includes pagination metadata.

```
{
  "success": true,
  "message": "Request completed successfully",
  "data": [
    // ...
  ],
  "meta": {
    "pagination": {
      "page": 1,
      "limit": 10,
      "totalItems": 125,
      "totalPages": 13,
      "hasNextPage": true,
      "hasPreviousPage": false
    }
  }
}
```

7.3. Error Response

All error responses also follow the same envelope and include an `error` object.

```
{
  "success": false,
  "message": "Validation failed",
  "data": null,
  "meta": null,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "The uploaded file is missing the required column:
      gender."
  }
}
```

7.4. Health Check

Endpoint

GET /health

Description

Checks whether the API is running.

Successful Response

```
{
  "success": true,
  "message": "Request completed successfully",
  "data": {
    "status": "ok",
    "service": "Civara API"
  },
  "meta": null
}
```

Status Codes

Status Code	Meaning
200	API is running successfully.

Table 2: Health check status code

7.5. Analyze Dataset

Endpoint

```
POST /analyze
```

Description

Uploads a CSV dataset and returns statistical analysis results.

Request Type

```
multipart/form-data
```

Request Body

Field	Type	Description
file	File	The CSV dataset to be analyzed.
scoreType	String	Optional. Defaults to average.

Table 3: Analyze dataset request body

Supported Score Types

Value	Description
average	Uses the average of math, reading, and writing scores.
math	Uses math score only.
reading	Uses reading score only.
writing	Uses writing score only.

Table 4: Supported score types

8. Successful Analysis Response

Status Code

```
200 OK
```

Response Body

```
{
```

```

"success": true,
"message": "Request completed successfully",
"data": {
  "project": "Civara",
  "analysis": {
    "scoreType": "average",
    "groupVariable": "gender",
    "groups": {
      "male": {
        "sampleSize": 482,
        "mean": 65.84,
        "standardDeviation": 14.10,
        "minimum": 9.00,
        "maximum": 100.00
      },
      "female": {
        "sampleSize": 518,
        "mean": 69.57,
        "standardDeviation": 14.54,
        "minimum": 23.00,
        "maximum": 100.00
      }
    },
    "comparison": {
      "meanDifference": -3.73,
      "standardError": 0.91,
      "degreesOfFreedom": 996.12,
      "confidenceLevel": 0.95,
      "confidenceInterval": {
        "lower": -5.52,
        "upper": -1.94
      },
      "tStatistic": -4.09,
      "pValue": 0.00005,
      "isSignificant": true
    },
    "interpretation": "The 95% confidence interval does not
      contain zero. Therefore, there is a statistically
      significant difference between the mean scores of male and
      female students at the 5% significance level."
  }
}

```

```

},
"meta": null
}

```

9. Response Fields

Field	Description
success	Indicates whether the request was processed successfully.
message	A human-readable summary of the result.
data	The response payload. When success=true, this contains the expected result object or list.
meta	Optional metadata object. null when not applicable.
meta.pagination	Present only for list endpoints that support pagination.
data.project	The project name.
data.analysis.scoreType	The score used in the analysis.
data.analysis.groupVariable	The variable used to separate the two groups.
sampleSize	Number of observations in each group.
mean	Sample mean for each group.
standardDeviation	Sample standard deviation for each group.
minimum	Minimum score in each group.
maximum	Maximum score in each group.
meanDifference	Difference between the two group means.
standardError	Standard error of the difference between means.
degreesOfFreedom	Welch degrees of freedom.
confidenceLevel	Confidence level used in the interval.
confidenceInterval.lower	Lower bound of the confidence interval.
confidenceInterval.upper	Upper bound of the confidence interval.
tStatistic	Test statistic from Welch's two-sample t-test.
pValue	Probability value from the hypothesis test.
isSignificant	Indicates whether the result is statistically significant.
interpretation	Plain-language statistical conclusion.

Table 5: Analysis response fields

10. Statistical Method

The API compares two independent groups using Welch's two-sample method.

The parameter of interest is:

$$\mu_{\text{male}} - \mu_{\text{female}}$$

The standard error is calculated as:

$$SE = \sqrt{\frac{s_m^2}{n_m} + \frac{s_f^2}{n_f}}$$

The 95% confidence interval is calculated as:

$$(\bar{x}_m - \bar{x}_f) \pm t_{\alpha/2} \times SE$$

where:

- $\bar{x}_m$ is the male sample mean.
- $\bar{x}_f$ is the female sample mean.
- s_m is the male sample standard deviation.
- s_f is the female sample standard deviation.
- n_m is the male sample size.
- n_f is the female sample size.

The null and alternative hypotheses are:

$$H_0 : \mu_{\text{male}} = \mu_{\text{female}}$$

$$H_1 : \mu_{\text{male}} \neq \mu_{\text{female}}$$

The result is considered statistically significant when:

$$p < 0.05$$

11. Validation Rules

The API validates the uploaded dataset before analysis.

Validation Rule	Expected Behavior
Missing file	Return a validation error.
Invalid file type	Return a validation error.
Missing required columns	Return a validation error.
Non-numeric score values	Return a validation error.
Less than two gender groups	Return a validation error.
Empty dataset	Return a validation error.

Table 6: Dataset validation rules

12. Error Response Format

All error responses follow the standard response envelope.

```
{
  "success": false,
  "message": "Validation failed",
  "data": null,
  "meta": null,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "The uploaded file is missing the required column:
                gender."
  }
}
```

Error Code	HTTP Status	Description
NO_FILE	400	No file was uploaded.
INVALID_FILE_TYPE	400	The uploaded file is not a CSV file.
MISSING_COLUMNS	400	One or more required columns are missing.
INVALID_DATA	400	The dataset contains invalid or non-numeric values.
INSUFFICIENT_GROUPS	400	The dataset does not contain two valid groups.
ANALYSIS_FAILED	500	The server failed to complete the analysis.

Table 7: API error codes

13. Error Codes

14. Example Request

Using curl:

```
curl -X POST https://civara.mahros.dev/api/v1/analyze \
-F "file=@students.csv" \
-F "scoreType=average"
```

15. Client-Side Cache Recommendation

Since the API does not use a database, the frontend may store the latest result in browser storage.

Recommended browser storage key:

```
civara:last-analysis
```

The cached object may contain:

```
{
  "uploadedAt": "2026-05-12T10:30:00.000Z",
  "fileName": "students.csv",
  "analysis": {}
}
```

This allows the dashboard to reload the latest analysis without requiring the user to upload the dataset again.

16. Conclusion

This API contract defines a minimal and practical public interface for Civara. The design is intentionally simple: a user uploads a CSV dataset, the API validates and analyzes the data, and the response contains the statistical results needed for interpretation.

The API supports the main goal of the project by converting a manual statistical workflow into a small, usable software system.